

Transition-Metal-Free Site-Selective β - and γ -C–H Borylation of Aliphatic AminesSumon Sarkar,[#] Nitish Kumar Deo,[#] Tong Zhang, and Vladimir Gevorgyan*Cite This: *J. Am. Chem. Soc.* 2025, 147, 40064–40070

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ABSTRACT: A mild photoinduced transition metal-free method for β - and γ -C(sp³)–H borylation of amines has been developed. This protocol features a regioselective intramolecular hydrogen atom transfer (HAT) process to access key amino alkyl radical intermediate employing commercially available, installable/removable iodoaryl tethers. Notably, this method introduces a new iodopyridyl-based hydrogen atom transferring group for abstraction of previously elusive β -C–H bonds of amines. Remarkably, this general and operationally simple method allows activation of primary, secondary, and tertiary C–H sites of a broad range of acyclic and cyclic secondary amines and offers a complementary platform toward regioselective and diastereoselective syntheses of valuable β - and γ -amino boronates. Utility of this protocol has been demonstrated through formal C–H oxidation reaction and other late-stage derivatizations of amines.

Organoboron compounds play a vital role in a multitude of areas, ranging from organic synthesis to medicinal chemistry and material science.¹ Specifically, aminoboronic acids act as bioisosteres of native amino acids and serve as a pivotal motif in drugs and bioactive molecules.² Therefore, not surprisingly, significant advancements have been made in the synthesis of α -aminoboronic acids.^{3,4} The latter, however, suffer from low metabolic stability, which could be addressed by incorporating an additional carbon to form β - or γ -aminoboronates.⁵ Consequently, β - and γ -aminoboronic acids have emerged as promising candidates in medicinal chemistry, because of their distinct bioactivities (Scheme 1A).⁶ Additionally, β - and γ -aminoboronates serve as versatile intermediates for rapid synthesis of functionalized motifs, including β - and γ -amino alcohols, moieties commonly found in natural products and pharmaceuticals.⁷ Traditionally, synthesis of amino-boronates follows multistep routes⁸ or transition-metal-catalyzed reactions of functionalized alkenes,⁹ imines,^{6b} or aziridines.¹⁰ Undoubtedly, a more attractive approach would involve site-selective C(sp³)–H borylation of aliphatic amines.¹¹ The existing remote C(sp³)–H borylation approaches typically involve transition-metal-catalyzed chelation-assisted C–H activation, which is limited to functionalization of γ -primary¹² or acidic β -cyclopropyl C–H sites¹³ of primary amines (Scheme 1B). In these cases, the regioselectivity is governed by the thermodynamic stability of the metallacycle and formation of stronger carbon–metal bonds (“A” in Scheme 1B). Thus, activation of acyclic tertiary and secondary β - and γ -C–H sites in the presence of accessible primary C–H site remains unattainable.¹⁴ Another inherent limitation of this approach is the difficulties associated with activation of the hydridic C–H sites. Thus, a broadly applicable β - and γ -C–H borylation strategy, especially for secondary amines under mild conditions would be highly desirable. Herein, we report a transition-metal-free, remote β -

and γ -C–H borylation protocol of feedstock amines under operationally simple, photoinduced conditions (Scheme 1D).

In recent years, metal-free borylation reactions employing diboron reagents as radical trapping agents have gained significant traction in organic synthesis, which includes borylation of alkyl electrophiles, such as alkyl halide, as well as unusual primary selective C(sp³)–H borylation of aliphatic systems (Scheme 1C).¹⁵ In contrast to transition-metal-catalyzed C–H functionalizations, which are limited to primary C–H sites, radical C–H functionalization of amines proceeds via HAT, which prefers weaker secondary and tertiary C–H sites.¹⁶ However, current advancements in C(sp³)–H borylation of amines are limited to activation of weak α -C–H^{17,4} and remote δ -C–H¹⁸ sites due to innate preference of the radical translocating groups (Scheme 1C). In the past decade, many advancements on using removable tethers to achieve intramolecular site-selective HAT at unactivated C(sp³)–H sites for various transformations have been made.¹⁹ Yet, none of the established tethers have proven to be effective for HAT at the β -C–H sites of amines, especially in the presence of weak γ -C–H sites. We envisioned that commercially available 3-iodopyridyl-tethered amines could enable the elusive selective activation of secondary β -C–H bonds via 1,5-HAT mediated by the corresponding pyridyl radical (“B” in Scheme 1D). For synthesis of γ -aminoboronate, we thought to use 2-iodobenzenesulfonyl tethers, which are known to favor 1,7-HAT at the γ -C–H sites of aliphatic amines.²⁰ Potential pitfalls include premature borylation of

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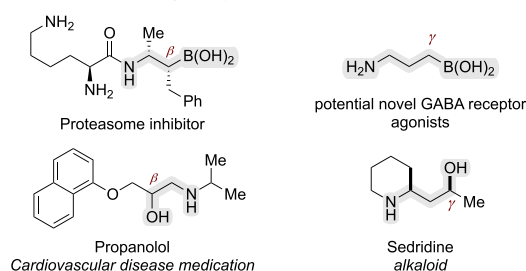
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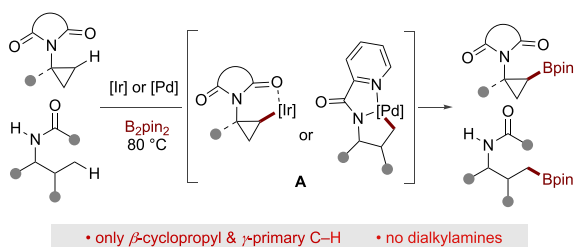


Scheme 1. Directed β - and γ -C(sp³)-H Borylation of Amines

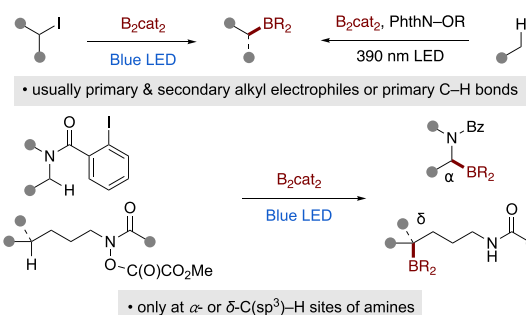
A. Representative examples of β - and γ -amino boronic acids and their derivatives



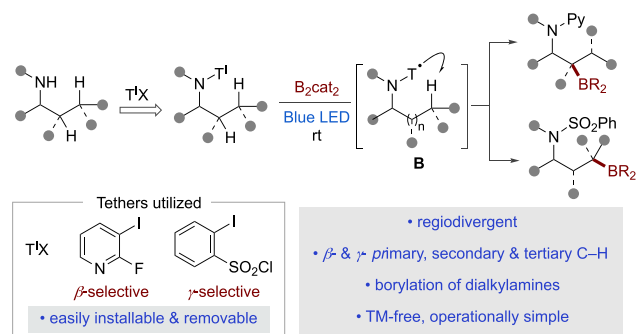
B. Transition metal (TM)-catalyzed β - and γ -C(sp³)-H borylation of amines



C. Selected Examples of TM-free diboron reagent-mediated aliphatic borylation



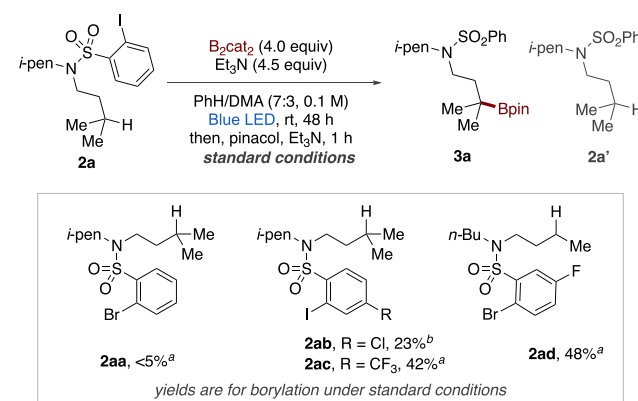
D. This work: TM-free regiodivergent β - and γ -selective C(sp³)-H borylation of amines



more-reactive aryl radical, cyclization or desaturation of the translocated alkyl radical, and hydrodehalogenation via intermolecular HAT. Thus, despite recent development with aryl-radical-mediated remote C-H functionalizations, successful interception of these alkyl radicals ("B" in Scheme 1D) for intermolecular functionalization remains challenging.

Our studies commenced with γ -borylation of 2-iodobenzenesulfonyl-tethered diisopentylamine **2a**. Gratifyingly, reaction of 2-iodobenzenesulfonyl-tethered diisopentylamine **2a** successfully afforded the targeted γ -borylation product **3a** in 74% isolated yield.²¹ The method employs bis(catecholato)diboron (B₂cat₂, 4.0 equiv) under 456 nm light irradiation and involves a simple workup procedure upon reaction completion (Table 1, entry 1). Notably, reaction conditions based on our previously

Table 1. Optimization of Reaction Conditions for γ -Borylation



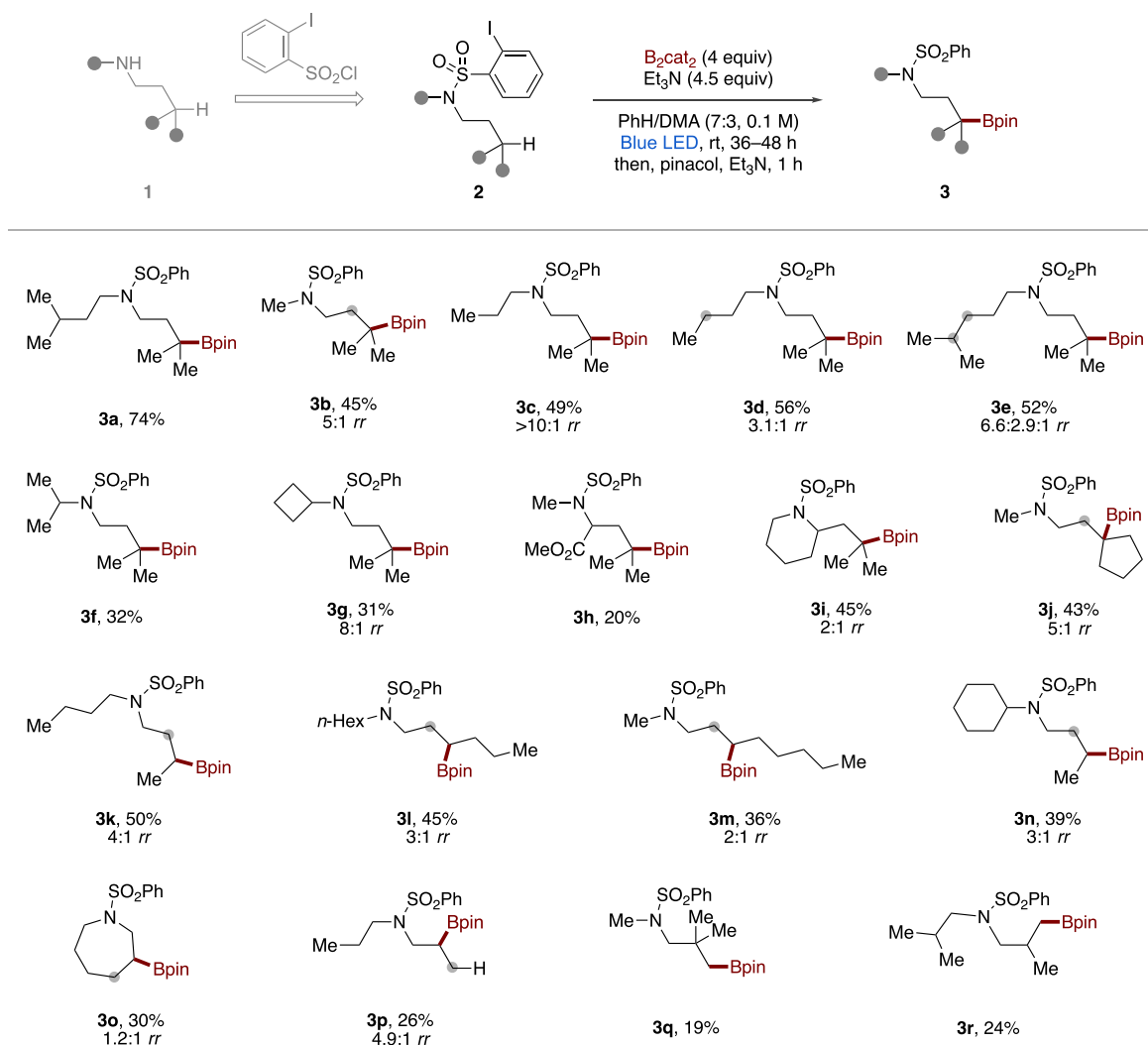
entry	deviation from standard conditions	yield of 3a ^a (%)
1	none	74 ^b
2	only DMA	45 ^c
3	only PhH	<5
4	B ₂ cat ₂ (2 equiv), Et ₃ N (2.5 equiv)	42
5	B ₂ pin ₂ as boron reagent	0
6	without Et ₃ N	16
7	under dark, 80 °C	<5

^a0.2 mmol scale, GC/MS or NMR yields. ^bIsolated yields.

^cHydrodehalogenation side product (**2a'**) formed (40%).

reported α -C-H borylation protocol,⁴ employing an amide solvent only, led to significantly diminished yields, likely due to the competing intramolecular HAT pathways that produced the reduction byproduct **2a'** (Table 1, entry 2). Performing the reaction in benzene alone failed to generate any γ -borylated product, underscoring the essential role of both cosolvents (Table 1, entry 3). The reaction remains viable with reduced B₂cat₂ loading, albeit with lower yields (Table 1, entry 4). Consistent with related reports (Table 1, entry 5),^{4,18} attempts to use more practical diboron reagent, such as bis(pinacolato)-diboron (B₂pin₂), were unsuccessful. Control experiments indicated no or inefficient reaction in the absence of triethylamine or under dark conditions (Table 1, entries 6 and 7). Screening *ortho*-halobenzenesulfonyl scaffolds (**2aa**–**2ad**) revealed no consistent electronic trend in reaction efficiency (Table 1).

With the optimized conditions in hand, we set out to explore the generality of the γ -borylation protocol (Scheme 2). Acyclic amines with γ -tertiary C-H afforded borylation products **3a** and **3b** in reasonable yields. However, borylation of the *N*-methyl analogue (**3b**), a substrate possessing a less bulky substituent at the *N*-atom, was less selective, also producing β -borylated product as a minor regioisomer. Expectedly, due to the lower bond dissociation energy and kinetic preference for the 1,7-HAT over the 1,6-HAT events, cyclic amines with competitive C-H sites, such as γ -primary, γ -secondary, and δ -tertiary C-H sites, led to borylation preferentially at the γ -tertiary sites (**3c**–**3e**). Of note, transition metal-catalyzed C-H activation methods would provide functionalization with complementary site selectivity, preferring least hindered and more acidic C-H sites. Next, the scope of *N*-alkyl substituted acyclic and cyclic amines was explored. A broad range of secondary amines (**3f**–**3j**), including the leucine derivative (**3h**), underwent site-selective borylation at the tertiary C-H sites of amines. Then, the possibility of a site-selective C-H

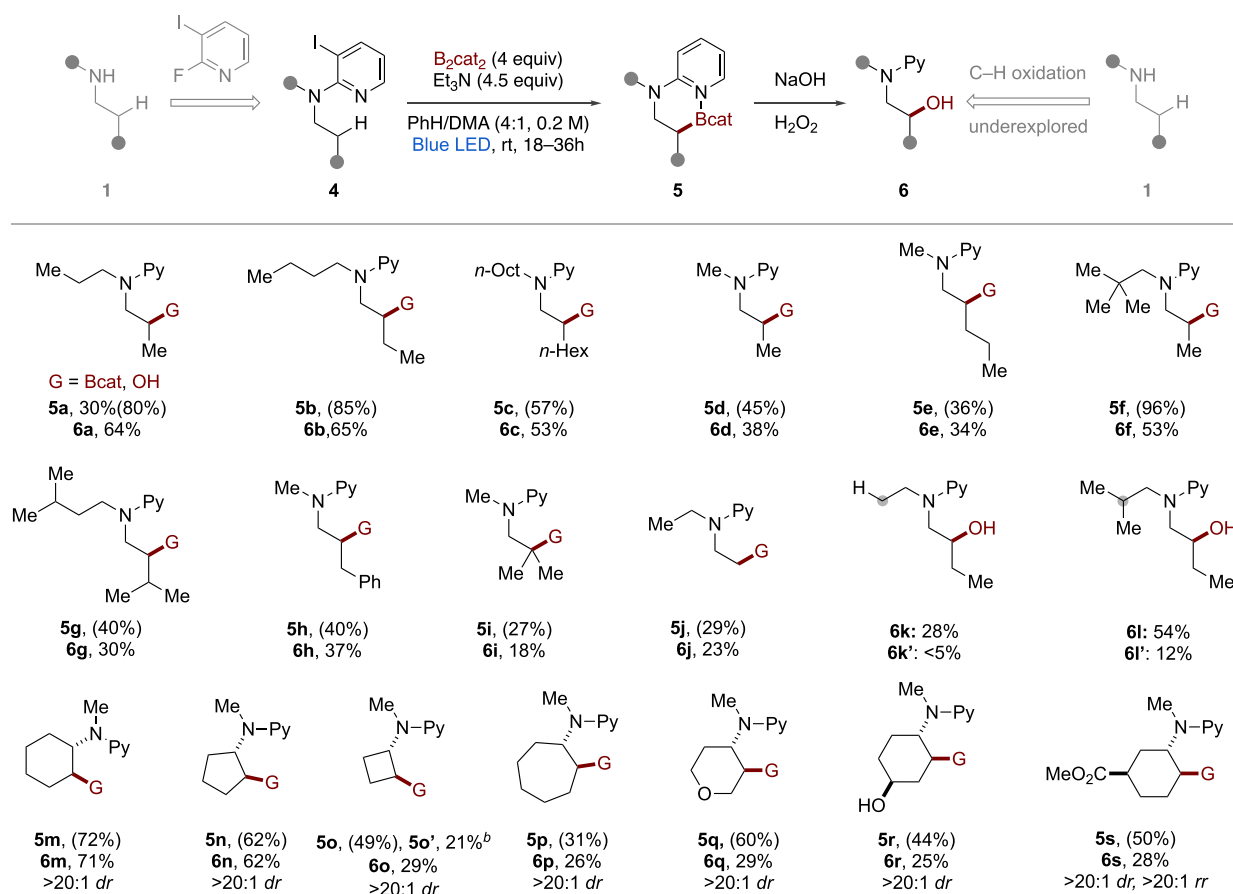
Scheme 2. Scope of γ -C–H Borylation of Amines^a

^a0.2 mmol scale, isolated yields, unless otherwise mentioned rr > 20:1.

borylation at the less reactive, and therefore more challenging, secondary C–H sites of acyclic substrates was examined. Gratifyingly, various acyclic amines (**3k–3n**) were selectively borylated under these conditions. The observed site selectivity was in agreement with the previously reported C–H activation reactivity patterns employing 2-iodosulfonyl tether.²⁰ Interestingly, cyclic amine azepane **3o** showed diminished γ -selectivity, delivering the β -borylation product as a slightly preferred regioisomer. Expectedly, in the absence of secondary γ -C–H sites, tethered dipropyl amine **3p** led to the β -borylation product. Remarkably, this protocol was even applicable to C–H borylation of unactivated primary $C(sp^3)$ –H sites, delivering aminoboronates **3q**, **3r** with up to 24% yield! The lower efficiency likely results from the reduced reactivity, because of the higher C–H bond dissociation energies relative to that of secondary and tertiary C–H bonds.¹⁹ Of note, although the recent direct C–H borylation methods allows access to primary C–H borylations of acyclic amines, they often struggle with sterically hindered neopentyl C–H bonds²² present in **3q**.

After establishing the scope for γ -borylation, we set out to explore the possibility of β -borylation of aliphatic amines (Scheme 3). Since the existing approaches for C–H functionalization of amines via intramolecular HAT do not

allow for activation of β -C–H bonds, especially in the presence of accessible γ -C–H bonds, the development of a new tether was required. Inspired by Curran's work²³ on use of *o*-bromo-*p*-methoxyphenyl ethers as radical translocating groups for β -C–H bond activation of ether, we sought to test iodoaryl tethered amines for C–H borylation. To our delight, 3-iodopyridyl tethered propyl amine **6a** successfully afforded the targeted β -C–H borylation product in 80% NMR yields.²¹ Like γ -C–H borylation, when 3-halopyridyl tethers with varying electronic and steric properties were evaluated, no clear correlation between electronic or steric effects and overall borylation yield was observed.²¹ Based on reactivity, commercial availability, cost, and ease of installation, 2-fluoro-3-iodopyridine was identified as the most practical and efficient directing group for this transformation. For ease of isolation, formed catechol boronates were oxidized into 1,2-aminoalcohols, which themselves are valuable synthetic motifs. Of note, synthesis of 1,2-aminoalcohols via formal or direct C–H oxidation of amines typically involve transition-metal-catalyzed chelation-assisted C–H activation, thus, usually limited to activation of primary C–H sites.²⁴ The scope of β -C–H borylation was found to be broad. Thus, symmetrical and *N*-alkylated amines underwent highly regioselective β -

Scheme 3. Scope of β -C–H Borylation Amines^a

^a0.2 mmol scale, isolated yields, NMR yields are in parentheses, unless otherwise mentioned $rr > 20:1$. ^bG = Bpin. After transesterification with pinacol.

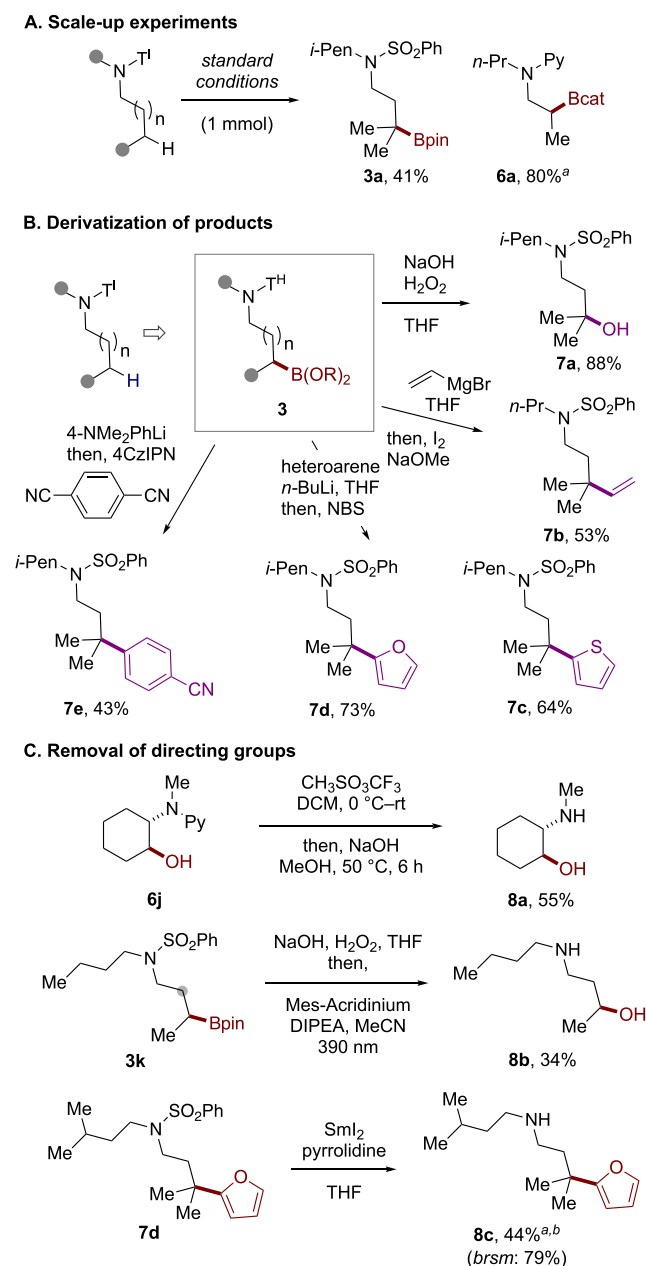
borylation, affording diverse aminoalcohols (**6a–6f**) with up to 65% overall yields. Expectedly, less bulky *N*-alkyl groups (**6d**, **6e**) resulted in diminished yields. The directed nature of this transformation allowed for highly regioselective β -C–H borylation at unactivated secondary sites, even in the presence of weaker γ -tertiary (**6g**) and benzylic (**6h**) sites. Notably, *N*-methyl isobutyl amine (**6i**) underwent β -selective borylation at the sterically hindered tertiary C–H sites, albeit with lower yields. Of note, no β -tertiary C–H borylation was observed in the case of the analogous sulfonyl tethered isobutyl amine substrate **3r**. Importantly, the pyridyl tether can also access unactivated β -primary C–H sites, affording **6j** in reduced yield. The lower efficiency likely results from the less-efficient 1,5-HAT step and competing side reactions, including aromatic ring borylation and hydrodehalogenation. While, transition-metal-catalyzed C–H oxidation of amines can access primary C–H sites, it is usually limited to primary amines.²⁴ In the case of amines with competitive primary, secondary, and tertiary C–H sites (**6k**, **6l**), a preference toward secondary C–H sites was observed, which could be attributed to a combination of BDEs and steric reasons. Next, *N*-methyl cyclic amines with various ring sizes were tested in this borylation reaction, which led to cyclic 1,2-aminols **6m–6s** in moderate to good yields with excellent diastereoselectivities. In the cases of **6n–6p**, increased conformational constraints likely hinder the formation of the quasi-linear transition state required for the 1,5-HAT event. Due to inefficient HAT, the competing side reactions, such as hydrodehalogenation, lead to the diminished

yields.¹⁹ This method is tolerated by functional groups, including free alcohol (**6r**) and ester (**6s**), and shows high site selectivity toward relatively more hydridic C–H sites (**6s**). Of note, the approach to forming some cyclic 1,2-aminols relevant to medicinal chemistry, such as the 2-(methylamino)-cyclobutan-1-ol derivative (**6o**), via conventional methods, would require multistep synthesis.²⁵

The practicality of this approach was demonstrated by the scale-up experiments with a lower equivalent of diboron reagents, to afford γ -borylation product **3a** and β -borylation product **6a** in good yields (Scheme 4A). In addition to oxidation to aminoalcohol (**7a**), the obtained aminobornates (**3**) could be vinyllated (**7b**), or (hetero)arylated (**7c–7e**) (Scheme 4B). Of note, it is highly challenging to access **7b–7e** via transition-metal-catalyzed direct C–H vinylation or arylation at the tertiary C–H sites of amines.^{20d} Moreover, it was also demonstrated that the pyridyl and benzenesulfonyl tethers could efficiently be removed to afford β -amino alcohol **8a**, γ -amino alcohol **8b**, and γ -heteroarylated amine **8c** respectively (Scheme 4C).

The following studies were performed to elucidate the mechanism of this transformation. First, employment of TEMPO and other radical scavengers²¹ led to suppression of borylation reaction (**4a** → **5a**), although with no trapping adduct **9a** observed (Scheme 5A). The radical nature of the transformation was supported by a radical probe experiment with **4t**, which produced the ring-opening borylation product **5t** (Scheme 5B). The quantum yield measurement ($\phi = 3.1$)

Scheme 4. Synthetic Utility

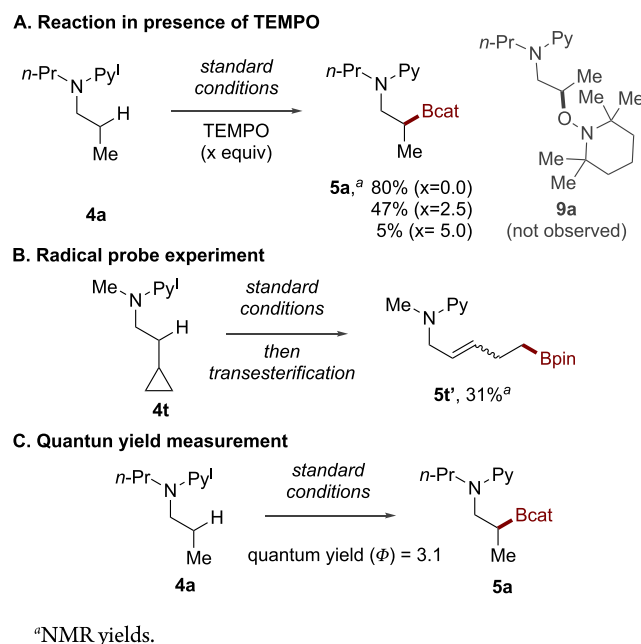


^aNMR yields. ^bIsolated after benzoyl protection.

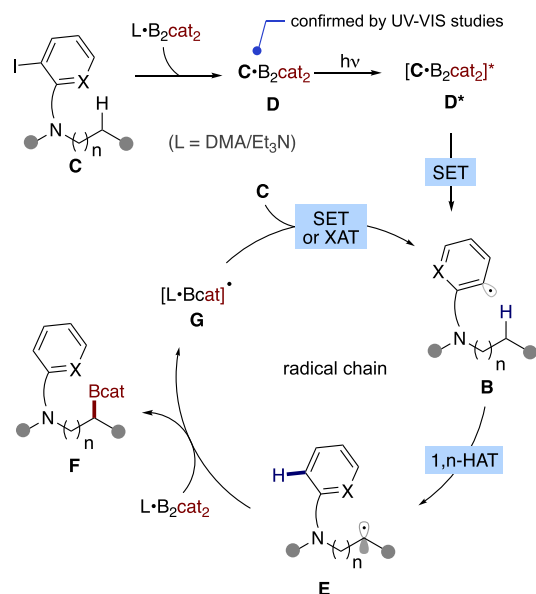
indicated the involvement of the radical chain mechanism (Scheme 5C).

Based on the preliminary mechanistic studies²¹ and literature reports,^{4,15} the following mechanism is proposed (Scheme 6). First, the tethered amine **C** forms an electron donor–acceptor (EDA) complex **D** with the diboron reagent, as supported by the results of the UV–vis studies. The latter, upon sequential photoexcitation (**D**^{*}), single-electron transfer (SET), and fragmentation of the C–I bond leads to an electrophilic aryl(hetaryl) radical **B**. A subsequent polarity-matched²⁶ 1,*n*-HAT (*n* = 5, 6, 7) to the latter from β- or γ-C–H site of alkyl amine generates a translocated nucleophilic alkyl radical **E**. A radical addition of it to the ligated diboron reagent affords borylated product **F**, along with boryl radical **G**, which propagates the radical chain via halogen-atom transfer (XAT) or SET with substrate **C**.

Scheme 5. Mechanistic Studies



Scheme 6. Proposed Mechanism



In summary, we developed a general transition-metal-free strategy for site-selective β- and γ-C(sp³)–H borylation of aliphatic secondary amines. This protocol involves photo-induced diboron reagent-mediated mild generation of an aryl radical and guided radical relay process to access the transposed alkyl radical employing commercially available, installable/removeable iodoaryl tethers. Notably, this work introduces 3-iodopyridyl group as a new radical translocating group for selective activation of previously inaccessible β-secondary C–H bonds of amines. This method, which does not require the use of photosensitizer, transition-metal catalyst, or harsh reagents, overcomes longstanding limitations of classical chelation-controlled or direct borylation approaches, particularly toward borylation at β- or γ-secondary and tertiary C–H sites of amines. This highly site-selective and diastereoselective method offers a complementary platform

toward efficient, site-selective C–H borylation of broad range of secondary amines, further enabling formal C–H oxidation and late-stage derivatizations.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.5c12618>.

Additional experimental details, materials, methods, and characterization data for all new compounds (PDF)

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Author Contributions

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Notes

The authors declare no competing financial interest.

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